



Lung Cancer Prevention (PDQ®)–Health Professional Version

[Go to Patient Version](#)

Overview

Note: The Overview section summarizes the published evidence on this topic. The rest of the summary describes the evidence in more detail.

Other PDQ summaries containing information related to lung cancer prevention include the following:

- [Lung Cancer Screening](#)
- [Small Cell Lung Cancer Treatment](#)
- [Non-Small Cell Lung Cancer Treatment](#)
- [Cigarette Smoking: Health Risks and How to Quit](#)

Who Is at Risk?

Lung cancer risk is largely a function of older age combined with extensive cigarette smoking history. Lung cancer is more common in men than women and in those of lower socioeconomic status. Patterns of lung cancer according to demographic characteristics tend to be strongly correlated with historical patterns of cigarette smoking prevalence. An exception is the very high rate of lung cancer in African American men, a group whose very high lung cancer death rate is not explainable simply by historical smoking patterns.^[1]

In nonsmokers, important lung cancer risk factors are exposure to secondhand smoke, exposure to ionizing radiation, and occupational exposure to lung carcinogens, such as asbestos. Radiation exposures relevant to the general population include environmental exposure to radon and radiation exposures administered in the medical care setting, particularly when administered at high doses, such as radiation therapy to the chest or breast. ^[2] Cigarette smoking often interacts with these other factors. There are several examples, including radon exposure and asbestos exposure, in which the combined exposure to cigarette smoke plus another risk factor results in an increase in risk that is much greater than the sum of the risks associated with each factor alone.

Factors associated with increased risk of lung cancer

Cigarette smoking

Starting with the 1964 Surgeon General's Report and followed by each subsequent Surgeon General's Report that has included a review of the evidence on smoking and lung cancer, an enormous body of scientific evidence clearly documents that cigarette smoking causes lung cancer, and that cigarette smoking is the major cause of lung cancer.

Based on solid evidence, cigarette smoking causes lung cancer. The risks of lung cancer associated with cigarette smoking are dose-dependent and increase markedly according to the number of cigarettes smoked per day and the number of years smoked. On average, current smokers have approximately 20 times the risk of lung cancer as nonsmokers.

Magnitude of Effect: Increased risk, very large.

Study Design: Numerous prospective cohort and case-control studies, combined with quasi-experimental evidence showing population-level smoking prevalence predicts the population-level burden of lung cancer.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Exposure to secondhand smoke

Based on solid evidence, exposure to secondhand smoke is an established cause of lung cancer.

Magnitude of Effect: Increased risk, small magnitude. Compared with nonsmokers not exposed to secondhand smoke, nonsmokers exposed to secondhand smoke have approximately a 20% increased risk of lung cancer.

Study Design: Prospective cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Radiation exposure

Based on solid evidence, exposure to radiation increases lung cancer incidence and mortality. Cigarette smoking greatly potentiates this effect.

Magnitude of Effect: Increased risk that follows a dose-response gradient, with smaller increases in risk for low levels of exposure and greater increases in risk for high levels of exposure.

Study Design: Cohort and case-control studies.

Internal Validity: Fair.

Consistency: Good.

External Validity: Good.

Occupational exposure to lung carcinogens

Based on solid evidence, workplace exposure to asbestos, arsenic, beryllium, cadmium, chromium, and nickel increases lung cancer incidence and mortality.

Magnitude of Effect: Increased risk, large magnitude (more than fivefold). Risks follow a dose-response gradient, with high-level exposures associated with large increases in risk. Cigarette smoking also potentiates the effect of many of these lung carcinogens so that the risks are even greater in smokers.

Study Design: Cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Air pollution

Based on solid evidence, exposure to outdoor air pollution, specifically small particles, increases lung cancer incidence and mortality.

Magnitude of Effect: Increased risk; compared with the lowest exposure categories, those in the highest exposure categories have approximately a 40% increased risk of lung cancer.

Study Design: Cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Factors of uncertain association with risk

Dietary factors

Based on equivocal evidence, the observed inverse associations between lung cancer and dietary factors, such as fruit and vegetable consumption, are difficult to disentangle from cigarette smoking.

Magnitude of Effect: Inverse association, moderate magnitude, but difficult to determine if true cause-effect association or due to confounding by cigarette smoking.

Study Design: Numerous cohort and case-control studies, and meta-analyses.

Internal Validity: Fair.

Consistency: Fair.

External Validity: Good.

Physical activity

Based on equivocal evidence, the observed inverse associations between lung cancer and higher levels of physical activity are difficult to disentangle from cigarette smoking.

Magnitude of Effect: Inverse association, moderate magnitude, but difficult to determine if true cause-effect association or due to confounding by cigarette smoking.

Study Design: Numerous cohort and case-control studies, and meta-analyses.

Internal Validity: Fair.

Consistency: Fair.

External Validity: Good.

Interventions Associated With Decreased Risk of Lung Cancer

Smoking avoidance

Based on solid evidence, cigarette smoking causes lung cancer and therefore, smoking avoidance results in decreased mortality from primary lung cancers.

Magnitude of Effect: Decreased risk, substantial magnitude.

Study Design: Cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Smoking cessation

Based on solid evidence, long-term sustained smoking cessation results in decreased incidence of lung cancer and of second primary lung tumors.

Magnitude of Effect: Decreased risk, moderate magnitude.

Study Design: Cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Eliminating secondhand smoke

Based on solid evidence, exposure to secondhand smoke causes lung cancer and therefore, preventing exposure to secondhand smoke results in decreased incidence and mortality from primary lung cancers.

Magnitude of Effect: Decreased risk, small magnitude.

Study Design: Cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Reducing or eliminating occupational exposure to lung carcinogens

Based on solid evidence, occupational exposures such as asbestos, arsenic, nickel, and chromium are causally associated with lung cancer. Reducing or eliminating workplace exposures to known lung carcinogens would be expected to result in a corresponding decrease in the risk of lung cancer.

Magnitude of Effect: Decreased risk, with a larger effect, the greater the reduction in exposure.

Study Design: Cohort and case-control studies.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Reducing or eliminating exposure to radon

Based on solid evidence, indoor exposure to radon increases lung cancer incidence and mortality, particularly among cigarette smokers. In homes with high radon concentrations, taking steps to prevent radon from entering homes by sealing the basement would be expected to result in a corresponding decrease in the risk of lung cancer.

Magnitude of Effect: Increased risk that follows a dose-response gradient, with small increases in risk for levels experienced in most at-risk homes to greater increases in risk for high-level exposures.

Study Design: Cohort and case-control studies.

Internal Validity: Fair.

Consistency: Good.

External Validity: Fair.

Interventions Associated With an Increased Risk of Lung Cancer

Beta-carotene supplementation in current smokers

Based on solid evidence, high-intensity smokers who take pharmacological doses of beta-carotene have an increased lung cancer incidence and mortality that is associated with taking the supplement.

Magnitude of Effect: Increased risk, small magnitude.

Study Design: Two randomized controlled trials (RCTs) with consistent results.

Internal Validity: Good.

Consistency: Good.

External Validity: Good.

Interventions That Do Not Decrease Risk of Lung Cancer

Beta-carotene in nonsmokers

Based on solid evidence, nonsmokers who take pharmacological doses of beta-carotene do not experience significantly different lung cancer incidence or mortality compared with taking a placebo.

Magnitude of Effect: No substantive effect.

Study Design: RCT.

Internal Validity: Good.

Consistency: Good.

External Validity: Fair.

Vitamin E (Tocopherol)

Based on solid evidence, taking vitamin E supplements does not affect the risk of lung cancer.

Magnitude of Effect: Strong evidence of no association.

Study Design: RCTs.

Internal Validity: Good.

Consistency: Fair.

External Validity: Good.

References

1. Pinsky PF: Racial and ethnic differences in lung cancer incidence: how much is explained by differences in smoking patterns? (United States). *Cancer Causes Control* 17 (8): 1017-24, 2006. [\[PUBMED Abstract\]](#)
2. Friedman DL, Whitton J, Leisenring W, et al.: Subsequent neoplasms in 5-year survivors of childhood cancer: the Childhood Cancer Survivor Study. *J Natl Cancer Inst* 102 (14): 1083-95, 2010. [\[PUBMED Abstract\]](#)

Incidence and Mortality

Lung cancer has a tremendous impact on the health of the American public, with an estimated 226,650 new cases and 124,730 deaths predicted in 2025 in men and women combined.[1] Lung cancer incidence and mortality rates increased markedly throughout most of the 20th century, first in men and then in women. The trends in lung cancer incidence and mortality rates have closely mirrored historical patterns of smoking prevalence, after accounting for an appropriate latency period. Because of historical differences in smoking prevalence between men and women, lung cancer rates in men have consistently declined since the mid-1980s, but rates in women have only declined since the mid-2000s.[2] The incidence rate in men declined from a high of 96.57 cases per 100,000 men in 1984 to 46.68 cases per 100,000 men in 2021. The incidence rate in women declined from a high of 51.01 cases per 100,000 women in 2005 to 41.58 cases per 100,000 women in 2021.[3] In the United States, lung cancer will account for about 11% of new cancer cases and about 20% of all cancer deaths in 2025. Lung cancer is the leading cause of cancer deaths in both men and women. In 2025, an estimated 60,540 deaths

due to lung cancer will occur among U.S. women, compared with an estimated 42,170 deaths due to breast cancer.[1]

Lung cancer incidence and mortality are highest in Black men compared with other racial and ethnic groups in the United States.[4] Between 2017 and 2021, the incidence rate in Black men was higher than in White men (66.5 vs. 60.0 cases per 100,000 men, respectively), whereas among women, the incidence rate in Black women was lower than in White women (43.2 vs. 52.2 cases per 100,000 women, respectively). Similarly, between 2018 and 2022, the mortality rate in Black men was higher than in White men (46.7 vs. 41.2 deaths per 100,000 men, respectively), whereas the mortality rate in Black women was lower than in White women (25.9 vs. 31.0 deaths per 100,000 women, respectively).[4]

Surgical treatment or radiation therapy is the treatment of choice for early stages of cancer.[5] The overall 5-year relative survival rate from lung cancer was 26.7% from 2014 to 2020. Lung cancer survival is worse for men compared with women and for Black individuals compared with White individuals. For example, 5-year lung cancer survival rates were lower in Black men than in White men (20.1% vs. 22.6%, respectively) and lower in Black women than in White women (28.2% vs. 30.8%, respectively).[3]

The hypothesis has been proposed that women may be more susceptible than men to lung cancer caused by smoking. However, the results of studies that compared the association between smoking and lung cancer in men and women using appropriate comparisons do not support this hypothesis.[6]

The results of the Multi-Ethnic Cohort Study indicated that for a given degree of cigarette smoking, African American individuals had a higher risk of lung cancer than other racial and ethnic groups.[7] Menthol cigarettes have been hypothesized as one potential factor contributing to the observed greater susceptibility to smoking-caused lung cancer in African American individuals, but menthol cigarettes have not been observed to be associated with a higher risk of lung cancer than nonmenthol cigarettes.[8,9]

References

1. American Cancer Society: Cancer Facts and Figures 2025. American Cancer Society, 2025. [Available online](#). Last accessed January 16, 2025.
2. Edwards BK, Brown ML, Wingo PA, et al.: Annual report to the nation on the status of cancer, 1975-2002, featuring population-based trends in cancer treatment. J Natl Cancer Inst 97 (19): 1407-27, 2005. [\[PUBMED Abstract\]](#)
3. Surveillance Research Program, National Cancer Institute: SEER*Explorer: An interactive website for SEER cancer statistics. Bethesda, MD: National Cancer Institute. [Available online](#). Last accessed December 30, 2024.

4. National Cancer Institute: SEER Stat Fact Sheets: Lung and Bronchus. Bethesda, Md: National Institutes of Health. [Available online](#). Last accessed December 12, 2024.
5. Spira A, Ettinger DS: Multidisciplinary management of lung cancer. N Engl J Med 350 (4): 379-92, 2004. [\[PUBMED Abstract\]](#)
6. Bain C, Feskanich D, Speizer FE, et al.: Lung cancer rates in men and women with comparable histories of smoking. J Natl Cancer Inst 96 (11): 826-34, 2004. [\[PUBMED Abstract\]](#)
7. Haiman CA, Stram DO, Wilkens LR, et al.: Ethnic and racial differences in the smoking-related risk of lung cancer. N Engl J Med 354 (4): 333-42, 2006. [\[PUBMED Abstract\]](#)
8. Blot WJ, Cohen SS, Aldrich M, et al.: Lung cancer risk among smokers of menthol cigarettes. J Natl Cancer Inst 103 (10): 810-6, 2011. [\[PUBMED Abstract\]](#)
9. Alberg AJ, Ford JG, Samet JM, et al.: Epidemiology of lung cancer: ACCP evidence-based clinical practice guidelines (2nd edition). Chest 132 (3 Suppl): 29S-55S, 2007. [\[PUBMED Abstract\]](#)

Cigarette Smoking Is the Primary Risk Factor

The epidemic of lung cancer in the 20th century was primarily due to increases in cigarette smoking, the predominant cause of lung cancer. The threefold variation in lung cancer mortality rates across the United States more or less parallels long-standing state-specific differences in the prevalence of cigarette smoking. For example, average annual age-adjusted lung cancer death rates for 1996 to 2000 were highest in Kentucky (78 deaths per 100,000 individuals), where 31% of residents were current smokers in 2001. Lung cancer death rates were lowest in Utah (26 deaths per 100,000 individuals), which had the lowest prevalence of cigarette smoking (13%).^[1]

References

1. Weir HK, Thun MJ, Hankey BF, et al.: Annual report to the nation on the status of cancer, 1975-2000, featuring the uses of surveillance data for cancer prevention and control. J Natl Cancer Inst 95 (17): 1276-99, 2003. [\[PUBMED Abstract\]](#)

The Biology of Carcinogenesis

Understanding the biology of carcinogenesis is crucial to the development of effective prevention and treatment strategies. Two important concepts in this regard are the multistep

nature of carcinogenesis and the diffuse field-wide carcinogenic process. Epithelial cancers in the lung appear to develop in a series of steps extending over years. Epithelial carcinogenesis is conceptually divided into three phases: initiation, promotion, and progression. This process has been inferred from human studies identifying clinical-histological premalignant lesions (e.g., metaplasia and dysplasia). The concept of field carcinogenesis is that multiple independent neoplastic lesions occurring within the lung can result from repeated exposure to carcinogens, primarily tobacco. Patients developing cancers of the aerodigestive tract secondary to cigarette smoke also are likely to have multiple premalignant lesions of independent origin within the carcinogen-exposed field. The concepts of multistep and field carcinogenesis provide a model for prevention studies.[1]

References

1. Lippman SM, Benner SE, Hong WK: Cancer chemoprevention. J Clin Oncol 12 (4): 851-73, 1994. [\[PUBMED Abstract\]](#)

Risk Factors

Factors Associated With Increased Risk of Lung Cancer

Cigarette smoking

The most important risk factor for lung cancer (and for many other cancers) is cigarette smoking.[1-3] Epidemiological data have established that cigarette smoking is the predominant cause of lung cancer. This causative link has been widely recognized since the 1960s, when national reports in Great Britain and the United States brought the cancer risk of smoking prominently to the public's attention.[2] The lifetime risk of lung cancer was estimated in a Swiss population to be 15% in men who smoke and 12% in women who smoke, compared with 2% or less in nonsmokers.[4] The percentages of lung cancers estimated to be caused by tobacco smoking in males and females are 90% and 78%, respectively. The manufactured cigarette has changed over time, but there is no evidence to suggest that changes such as *low tar* or *low nicotine* cigarettes have resulted in reduced lung cancer risks.[5,6] Cigarette smoking is the most important contributor to the lung cancer burden because of its high prevalence of use and because cigarette smokers tend to smoke frequently, but cigar and pipe smoking have also been associated independently in case-control and cohort studies with increased lung cancer risk.[7,8] The cigar risks are of particular concern because of the increased prevalence of cigar use in the United States.[9]

The development of lung cancer is the culmination of multistep carcinogenesis. Genetic damage caused by chronic exposure to carcinogens, such as those in cigarette smoke, is the driving force behind the multistep process. Evidence of genetic damage is the association of

cigarette smoking with the formation of the DNA adducts in human lung tissue. A strong link between tobacco smoke and lung carcinogenesis has been established by molecular data. [10,11]

Secondhand tobacco smoke

Secondhand tobacco smoke is also an established cause of lung cancer.[12] Secondhand smoke has the same components as inhaled mainstream smoke, though in lower absolute concentrations, between 1% and 10% depending on the constituent. Elevated biomarkers of tobacco exposure, including urinary cotinine, urinary 4-(methylnitrosamino)-1-(3-pyridyl)-1-butanone (NNK) metabolites, and carcinogen-protein adducts, are seen in those who are exposed to secondhand cigarette smoke.[13-15]

Family history

A positive family history of lung cancer is a risk factor for lung cancer. The results of a meta-analysis of epidemiological studies indicated that those with a positive family history of lung cancer were at approximately twice the risk of lung cancer compared with those with no affected relatives.[16,17] Cigarette smoking behavior tends to run in families and family members are exposed to secondhand smoke, so the extent to which measured family history represents a genetic predisposition to lung cancer independent of the shared risk factor of cigarette smoking is uncertain.

Human immunodeficiency virus (HIV) infection

HIV infection has been observed to be statistically associated with an increased lung cancer risk; for example, the results of a meta-analysis of 13 studies indicated HIV-infected individuals had a 2.6-fold higher risk of lung cancer than non-HIV-infected individuals (standard incidence ratio, 2.6; 95% confidence interval [CI], 2.1–3.1).[18] The clinical significance of this association remains to be elucidated, as it raises the possibility that HIV infection increases susceptibility to lung cancer, but may merely reflect the high smoking prevalence (study estimates ranged from 59% to 96%) among those infected with HIV compared with the general population (smoking prevalence approximately 20%).

Other environmental causes of lung cancer

Occupational exposures to lung carcinogens

Several environmental exposures other than tobacco smoke are causally associated with lung cancer, but the proportion of the lung cancer burden due to these exposures is small compared with cigarette smoking. Many lung carcinogens have been identified in studies of high occupational exposures. Considered in total, occupational exposures have been estimated to account for approximately 10% of lung cancers.[19] These carcinogens include asbestos, radon, tar and soot (sources of polycyclic aromatic hydrocarbons), arsenic, chromium, nickel,

beryllium, and cadmium.[20] For many of these workplace carcinogens, cigarette smoking interacts synergistically to increase the risk.[21] In developed countries, workplace exposures to these agents have largely been controlled.

Radiation exposure

Based on studies of populations exposed to high doses of radiation, lung cancer has been determined to be one of the cancers that is causally associated with exposure to ionizing radiation.[22] Two types of radiation that are relevant to lung cancer include high-energy ionizing electromagnetic radiation (such as x-rays and gamma rays) and particles (such as alpha particles and neutrons).

An important early source of data about radiation exposure came from studies of atomic bomb survivors in Japan; these studies demonstrated that a single high-dose exposure to gamma rays is sufficient to increase the risk of lung cancer in a dose-dependent fashion.[23] Lung cancer risk in patients treated with radiation for a number of medical conditions has also been evaluated. Studies of patients with tuberculosis who were treated with pneumothorax and monitored with frequent fluoroscopy, with resulting cumulative radiation doses of about 85 rads (0.85 Gy) staggered over time, indicated that any lung cancer risks associated with this exposure pattern, if they exist, are difficult to detect.[24,25] In contrast, the results of many studies provide clear-cut evidence that radiation therapy to the chest to treat cancer results in an increased risk of lung cancer in a dose-dependent manner. The evidence is most abundant for breast cancer [26-29] and Hodgkin lymphoma.[30] The risk of lung cancer after radiation therapy is amplified among patients who smoke cigarettes, compared with nonsmokers. [26,27,29,30]

The association between radiation exposure and lung cancer has implications for the general population in countries such as the United States, where computed tomography (CT) scans are relatively common and may contribute to an excess of cancer at the population level.[31] In light of the established association between exposure to ionizing radiation and lung cancer risk, researchers have urged caution to minimize risks when cancer screening involves ionizing radiation exposure, such as using of low-dose spiral CT screening for lung cancer instead of higher-dose techniques.[32,33]

Because they deposit concentrated energy in tissue, particles (e.g., alpha particles) produce more biological damage at an equivalent dose than radiation (e.g., x-rays).[34] A public health concern is radon, the primary source of alpha particles. Radon is an inert gas produced naturally in the decay series of uranium. Along with other supportive scientific evidence, studies of underground uranium miners exposed to very high levels of radon have demonstrated that radon exposure causes lung cancer.[22] This effect is amplified considerably in miners who smoke.[35] Radon has broader societal interest because it can enter buildings as a soil-derived gas and is a prevalent population-level exposure.

Estimates of the proportion of lung cancer deaths attributable to indoor exposure to radon vary by method of estimation and by the levels of radon exposure in a country, but the median estimates are 26% for lifelong nonsmokers (range, 13%–50%) and 10% for ever smokers (range, 7%–13%).[\[36-38\]](#) Because of a synergistic interaction between cigarette smoking and radon exposure, the radon-associated risk of lung cancer among smokers is considerably greater than for nonsmokers.[\[39\]](#) The prevention strategy for residents of homes with high radon concentrations is to have the basement sealed to prevent radon gas from leaking into the home.[\[40\]](#)

Air pollution

Although early evidence from case-control and cohort studies did not support an association between air pollution and lung cancer, the evidence now points to a genuine association.[\[41\]](#) In particular, two prospective cohort studies provide evidence to suggest that air pollution is weakly associated with the risk of lung cancer. In an extended follow-up of a study of six U.S. cities, the adjusted relative risk (RR) of lung cancer mortality for each 10 $\mu\text{g}/\text{m}^3$ increase in concentration of fine-particulate was 1.27 (95% CI, 0.96–1.69).[\[42\]](#) Using data from the American Cancer Society's Cancer Prevention Study II, it was observed that compared with the least polluted areas, residence in areas with high sulfate concentrations was associated with an increased risk of lung cancer (adjusted RR, 1.4; 95% CI, 1.1–1.7) after adjustment for occupational exposures and the factors mentioned above.[\[43\]](#) In a subsequent update to this report, the risk of lung cancer was observed to increase 14% for each 10 $\mu\text{g}/\text{m}^3$ increase in concentration of fine particles.[\[44\]](#) The evidence indicating an association between constituents of ambient air pollution and increased lung cancer mortality continues to strengthen, with reports from Asia,[\[45,46\]](#) New Zealand,[\[47\]](#) and Europe,[\[48\]](#) documenting increased risks with exposure to certain components of particulate matter.

Factors of Uncertain Association With Risk

Dietary factors

Studies of dietary factors have yielded intriguing findings, but because the diets of smokers tend to be less healthy than those of nonsmokers, it is challenging to separate the influence of dietary factors from the effects of smoking. When considering the relationships between lung cancer and dietary factors, confounding factors related to cigarette smoking cannot be dismissed as a possible explanation.

While the focus has been on fruit and vegetable consumption and micronutrients, a wide range of dietary and anthropometric factors have been investigated. Anthropometric measures have been studied, indicating a tendency for leaner persons to have increased lung cancer risk relative to those with greater body mass index.[\[49,50\]](#) The results of a meta-analysis showed

that alcohol drinking in the highest consumption categories only (in excess of about a drink a day) was associated with an increased risk of lung cancer.

Physical activity

A meta-analysis of leisure-time physical activity and lung cancer risk revealed that higher levels of physical activity protect against lung cancer.[51] The overall evidence for physical activity has been mixed, but several studies have reported that individuals who are more physically active have a lower risk of lung cancer than those who are more sedentary,[52-54] even after adjustment for cigarette smoking. The WCRF evidence review rated the inverse association between physical activity and lung cancer as *limited suggestive* evidence.[55]

Studies of physical activity yield findings consistent with an inverse association, but because physical activity behaviors differ between smokers and nonsmokers, it is difficult to infer that there is a direct relationship between physical activity and lung cancer risk.

Lung cancer in never smokers

In countries where cigarette smoking is common, about 10% to 20% of lung cancer cases occur in never smokers.[56] Radon and secondhand smoke exposure are established causes of lung cancer in never smokers. An increase in lung cancer risk among never smokers also has been observed with exposure to asbestos, ionizing radiation from sources other than radon, and indoor air pollution caused by combustion of coal or other solid fuel.[57] Limited data are available about the association of lung cancer in never smokers with physical activity, diet, alcohol, and anthropometry, yet they typically suggest that the relationships do not differ markedly from those in ever smokers.[49,52-54,58-60] Nevertheless, the inability to fully control for confounding by smoking in epidemiological analyses of ever smokers and the possibility of different lung cancer causal pathways from never and ever smokers warrants care when extrapolating results for ever smokers to never smokers.

References

1. Schottenfeld D, Fraumeni JF Jr, eds.: Cancer Epidemiology and Prevention. 2nd ed. Oxford University Press, 1996.
2. Smoking and Health: Report of the Advisory Committee to the Surgeon General of the Public Health Service. US Department of Health, Education, and Welfare, 1965. PHS Publ No 1103.
3. Gazdar AF, Minna JD: Cigarettes, sex, and lung adenocarcinoma. J Natl Cancer Inst 89 (21): 1563-5, 1997. [\[PUBMED Abstract\]](#)
4. Bruder C, Bulliard JL, Germann S, et al.: Estimating lifetime and 10-year risk of lung cancer. Prev Med Rep 11: 125-130, 2018. [\[PUBMED Abstract\]](#)

5. Song MA, Benowitz NL, Berman M, et al.: Cigarette Filter Ventilation and its Relationship to Increasing Rates of Lung Adenocarcinoma. *J Natl Cancer Inst* 109 (12): , 2017. [\[PUBMED Abstract\]](#)
6. Alberg AJ, Ford JG, Samet JM, et al.: Epidemiology of lung cancer: ACCP evidence-based clinical practice guidelines (2nd edition). *Chest* 132 (3 Suppl): 29S-55S, 2007. [\[PUBMED Abstract\]](#)
7. Iribarren C, Tekawa IS, Sidney S, et al.: Effect of cigar smoking on the risk of cardiovascular disease, chronic obstructive pulmonary disease, and cancer in men. *N Engl J Med* 340 (23): 1773-80, 1999. [\[PUBMED Abstract\]](#)
8. Boffetta P, Pershagen G, Jöckel KH, et al.: Cigar and pipe smoking and lung cancer risk: a multicenter study from Europe. *J Natl Cancer Inst* 91 (8): 697-701, 1999. [\[PUBMED Abstract\]](#)
9. Satcher D: Cigars and public health. *N Engl J Med* 340 (23): 1829-31, 1999. [\[PUBMED Abstract\]](#)
10. Mao L, Lee JS, Kurie JM, et al.: Clonal genetic alterations in the lungs of current and former smokers. *J Natl Cancer Inst* 89 (12): 857-62, 1997. [\[PUBMED Abstract\]](#)
11. Wistuba II, Lam S, Behrens C, et al.: Molecular damage in the bronchial epithelium of current and former smokers. *J Natl Cancer Inst* 89 (18): 1366-73, 1997. [\[PUBMED Abstract\]](#)
12. Hackshaw AK, Law MR, Wald NJ: The accumulated evidence on lung cancer and environmental tobacco smoke. *BMJ* 315 (7114): 980-8, 1997. [\[PUBMED Abstract\]](#)
13. Anderson KE, Carmella SG, Ye M, et al.: Metabolites of a tobacco-specific lung carcinogen in nonsmoking women exposed to environmental tobacco smoke. *J Natl Cancer Inst* 93 (5): 378-81, 2001. [\[PUBMED Abstract\]](#)
14. Anderson KE, Kliris J, Murphy L, et al.: Metabolites of a tobacco-specific lung carcinogen in nonsmoking casino patrons. *Cancer Epidemiol Biomarkers Prev* 12 (12): 1544-6, 2003. [\[PUBMED Abstract\]](#)
15. Tulunay OE, Hecht SS, Carmella SG, et al.: Urinary metabolites of a tobacco-specific lung carcinogen in nonsmoking hospitality workers. *Cancer Epidemiol Biomarkers Prev* 14 (5): 1283-6, 2005. [\[PUBMED Abstract\]](#)
16. Lissowska J, Foretova L, Dabek J, et al.: Family history and lung cancer risk: international multicentre case-control study in Eastern and Central Europe and meta-analyses. *Cancer Causes Control* 21 (7): 1091-104, 2010. [\[PUBMED Abstract\]](#)
17. Matakidou A, Eisen T, Houlston RS: Systematic review of the relationship between family history and lung cancer risk. *Br J Cancer* 93 (7): 825-33, 2005. [\[PUBMED Abstract\]](#)
18. Shiels MS, Cole SR, Kirk GD, et al.: A meta-analysis of the incidence of non-AIDS cancers in HIV-infected individuals. *J Acquir Immune Defic Syndr* 52 (5): 611-22, 2009. [\[PUBMED Abstract\]](#)

[Abstract\]](#)

19. Alberg AJ, Samet JM: Epidemiology of lung cancer. Chest 123 (1 Suppl): 21S-49S, 2003. [\[PUBMED Abstract\]](#)
20. Straif K, Benbrahim-Tallaa L, Baan R, et al.: A review of human carcinogens--part C: metals, arsenic, dusts, and fibres. Lancet Oncol 10 (5): 453-4, 2009. [\[PUBMED Abstract\]](#)
21. Saracci R: The interactions of tobacco smoking and other agents in cancer etiology. Epidemiol Rev 9: 175-93, 1987. [\[PUBMED Abstract\]](#)
22. Committee on Health Risks of Exposure to Radon (BEIR VI): Health Effects of Exposure to Radon: BEIR VI. National Academies Press, 1999. [Also available online](#). Last accessed December 12, 2024.
23. Shimizu Y, Kato H, Schull WJ: Studies of the mortality of A-bomb survivors. 9. Mortality, 1950-1985: Part 2. Cancer mortality based on the recently revised doses (DS86). Radiat Res 121 (2): 120-41, 1990. [\[PUBMED Abstract\]](#)
24. Davis FG, Boice JD, Hrubec Z, et al.: Cancer mortality in a radiation-exposed cohort of Massachusetts tuberculosis patients. Cancer Res 49 (21): 6130-6, 1989. [\[PUBMED Abstract\]](#)
25. Howe GR: Lung cancer mortality between 1950 and 1987 after exposure to fractionated moderate-dose-rate ionizing radiation in the Canadian fluoroscopy cohort study and a comparison with lung cancer mortality in the Atomic Bomb survivors study. Radiat Res 142 (3): 295-304, 1995. [\[PUBMED Abstract\]](#)
26. Lorigan P, Califano R, Faivre-Finn C, et al.: Lung cancer after treatment for breast cancer. Lancet Oncol 11 (12): 1184-92, 2010. [\[PUBMED Abstract\]](#)
27. Grantzau T, Thomsen MS, Væth M, et al.: Risk of second primary lung cancer in women after radiotherapy for breast cancer. Radiother Oncol 111 (3): 366-73, 2014. [\[PUBMED Abstract\]](#)
28. Grantzau T, Overgaard J: Risk of second non-breast cancer after radiotherapy for breast cancer: a systematic review and meta-analysis of 762,468 patients. Radiother Oncol 114 (1): 56-65, 2015. [\[PUBMED Abstract\]](#)
29. Kaufman EL, Jacobson JS, Hershman DL, et al.: Effect of breast cancer radiotherapy and cigarette smoking on risk of second primary lung cancer. J Clin Oncol 26 (3): 392-8, 2008. [\[PUBMED Abstract\]](#)
30. Lorigan P, Radford J, Howell A, et al.: Lung cancer after treatment for Hodgkin's lymphoma: a systematic review. Lancet Oncol 6 (10): 773-9, 2005. [\[PUBMED Abstract\]](#)
31. Alberg AJ, Brock MV, Ford JG, et al.: Epidemiology of lung cancer: Diagnosis and management of lung cancer, 3rd ed: American College of Chest Physicians evidence-based clinical practice guidelines. Chest 143 (5 Suppl): e1S-29S, 2013. [\[PUBMED Abstract\]](#)

32. Berrington de González A, Kim KP, Berg CD: Low-dose lung computed tomography screening before age 55: estimates of the mortality reduction required to outweigh the radiation-induced cancer risk. *J Med Screen* 15 (3): 153-8, 2008. [\[PUBMED Abstract\]](#)
33. Mascalchi M, Belli G, Zappa M, et al.: Risk-benefit analysis of X-ray exposure associated with lung cancer screening in the Italung-CT trial. *AJR Am J Roentgenol* 187 (2): 421-9, 2006. [\[PUBMED Abstract\]](#)
34. Hendee WR: Estimation of radiation risks. BEIR V and its significance for medicine. *JAMA* 268 (5): 620-4, 1992. [\[PUBMED Abstract\]](#)
35. Saccomanno G, Huth GC, Auerbach O, et al.: Relationship of radioactive radon daughters and cigarette smoking in the genesis of lung cancer in uranium miners. *Cancer* 62 (7): 1402-8, 1988. [\[PUBMED Abstract\]](#)
36. Kim SH, Hwang WJ, Cho JS, et al.: Attributable risk of lung cancer deaths due to indoor radon exposure. *Ann Occup Environ Med* 28: 8, 2016. [\[PUBMED Abstract\]](#)
37. Lubin JH, Boice JD: Lung cancer risk from residential radon: meta-analysis of eight epidemiologic studies. *J Natl Cancer Inst* 89 (1): 49-57, 1997. [\[PUBMED Abstract\]](#)
38. Krewski D, Lubin JH, Zielinski JM, et al.: Residential radon and risk of lung cancer: a combined analysis of 7 North American case-control studies. *Epidemiology* 16 (2): 137-45, 2005. [\[PUBMED Abstract\]](#)
39. Environmental Protection Agency: Exposure to Radon Causes Lung Cancer In Non-smokers and Smokers Alike. Washington, DC: Environmental Protection Agency, 2011. [Available online](#). Last accessed December 12, 2024.
40. Gray A, Read S, McGale P, et al.: Lung cancer deaths from indoor radon and the cost effectiveness and potential of policies to reduce them. *BMJ* 338: a3110, 2009. [\[PUBMED Abstract\]](#)
41. Vineis P, Forastiere F, Hoek G, et al.: Outdoor air pollution and lung cancer: recent epidemiologic evidence. *Int J Cancer* 111 (5): 647-52, 2004. [\[PUBMED Abstract\]](#)
42. Laden F, Schwartz J, Speizer FE, et al.: Reduction in fine particulate air pollution and mortality: Extended follow-up of the Harvard Six Cities study. *Am J Respir Crit Care Med* 173 (6): 667-72, 2006. [\[PUBMED Abstract\]](#)
43. Pope CA, Thun MJ, Namboodiri MM, et al.: Particulate air pollution as a predictor of mortality in a prospective study of U.S. adults. *Am J Respir Crit Care Med* 151 (3 Pt 1): 669-74, 1995. [\[PUBMED Abstract\]](#)
44. Pope CA, Burnett RT, Thun MJ, et al.: Lung cancer, cardiopulmonary mortality, and long-term exposure to fine particulate air pollution. *JAMA* 287 (9): 1132-41, 2002. [\[PUBMED Abstract\]](#)

45. Katanoda K, Sobue T, Satoh H, et al.: An association between long-term exposure to ambient air pollution and mortality from lung cancer and respiratory diseases in Japan. *J Epidemiol* 21 (2): 132-43, 2011. [\[PUBMED Abstract\]](#)
46. Cao J, Yang C, Li J, et al.: Association between long-term exposure to outdoor air pollution and mortality in China: a cohort study. *J Hazard Mater* 186 (2-3): 1594-600, 2011. [\[PUBMED Abstract\]](#)
47. Hales S, Blakely T, Woodward A: Air pollution and mortality in New Zealand: cohort study. *J Epidemiol Community Health* 66 (5): 468-73, 2012. [\[PUBMED Abstract\]](#)
48. Raaschou-Nielsen O, Andersen ZJ, Beelen R, et al.: Air pollution and lung cancer incidence in 17 European cohorts: prospective analyses from the European Study of Cohorts for Air Pollution Effects (ESCAPE). *Lancet Oncol* 14 (9): 813-22, 2013. [\[PUBMED Abstract\]](#)
49. Calle EE, Rodriguez C, Walker-Thurmond K, et al.: Overweight, obesity, and mortality from cancer in a prospectively studied cohort of U.S. adults. *N Engl J Med* 348 (17): 1625-38, 2003. [\[PUBMED Abstract\]](#)
50. Olson JE, Yang P, Schmitz K, et al.: Differential association of body mass index and fat distribution with three major histologic types of lung cancer: evidence from a cohort of older women. *Am J Epidemiol* 156 (7): 606-15, 2002. [\[PUBMED Abstract\]](#)
51. Tardon A, Lee WJ, Delgado-Rodriguez M, et al.: Leisure-time physical activity and lung cancer: a meta-analysis. *Cancer Causes Control* 16 (4): 389-97, 2005. [\[PUBMED Abstract\]](#)
52. Lee IM, Sesso HD, Paffenbarger RS: Physical activity and risk of lung cancer. *Int J Epidemiol* 28 (4): 620-5, 1999. [\[PUBMED Abstract\]](#)
53. Thune I, Lund E: The influence of physical activity on lung-cancer risk: A prospective study of 81,516 men and women. *Int J Cancer* 70 (1): 57-62, 1997. [\[PUBMED Abstract\]](#)
54. Mao Y, Pan S, Wen SW, et al.: Physical activity and the risk of lung cancer in Canada. *Am J Epidemiol* 158 (6): 564-75, 2003. [\[PUBMED Abstract\]](#)
55. Wiseman M: The second World Cancer Research Fund/American Institute for Cancer Research expert report. Food, nutrition, physical activity, and the prevention of cancer: a global perspective. *Proc Nutr Soc* 67 (3): 253-6, 2008. [\[PUBMED Abstract\]](#)
56. Peto R, Lopez AD, Boreham J, et al.: Mortality from Smoking in Developed Countries, 1950-2000: Indirect Estimates from National Vital Statistics. Oxford University Press, 1994.
57. Samet JM, Avila-Tang E, Boffetta P, et al.: Lung cancer in never smokers: clinical epidemiology and environmental risk factors. *Clin Cancer Res* 15 (18): 5626-45, 2009. [\[PUBMED Abstract\]](#)
58. Lam TK, Moore SC, Brinton LA, et al.: Anthropometric measures and physical activity and the risk of lung cancer in never-smokers: a prospective cohort study. *PLoS One* 8 (8): e70672, 2013. [\[PUBMED Abstract\]](#)

59. Gallicchio L, Boyd K, Matanoski G, et al.: Carotenoids and the risk of developing lung cancer: a systematic review. *Am J Clin Nutr* 88 (2): 372-83, 2008. [\[PUBMED Abstract\]](#)
60. Korte JE, Brennan P, Henley SJ, et al.: Dose-specific meta-analysis and sensitivity analysis of the relation between alcohol consumption and lung cancer risk. *Am J Epidemiol* 155 (6): 496-506, 2002. [\[PUBMED Abstract\]](#)

Interventions Associated With Decreased Risk of Lung Cancer

Smoking Avoidance and Cessation

Substantial harm to public health accrues from addiction to cigarette smoking. Compared with nonsmokers, smokers experience a dose-dependent increase in the risk of developing lung cancer (and many other malignancies).[1,2]

Approximately 85% of all lung cancer deaths are estimated to be attributed to cigarette smoking. Substantial benefits accrue to the smoker by quitting smoking. For more information, see [Cigarette Smoking: Health Risks and How to Quit](#). Avoidance of tobacco use is the most effective measure to prevent lung cancer. The preventive effect of smoking cessation depends on the duration and intensity of prior smoking and upon time since cessation. Compared with the risk in persistent smokers, a 30% to 60% reduction in lung cancer mortality risk has been noted after 10 years of cessation.[2-6] Although lung cancer mortality risk can be greatly reduced by quitting for a long period of time, the risk will never be as low as the risk in nonsmokers.[6] This emphasizes the importance of discouraging smoking initiation in younger people.

The benefits of tobacco control at the population level provide strong quasi-experimental evidence that reducing population-level exposure to cigarettes has resulted in population-level declines in the occurrence of lung cancer. Reduced tobacco consumption, resulting from decreases in smoking initiation and increases in smoking cessation, led to a decline in overall age-adjusted lung cancer mortality among men since the mid-1980s, consistent with reductions in smoking prevalence among men since the 1960s.[7] Gender differences in time trends for lung cancer are a reflection of (1) the later adoption of cigarette smoking in women compared with men and (2) the later reduction in smoking prevalence among women compared with men.

Smoking cessation guidelines

Nicotine dependence exposes smokers in a dose-dependent fashion to carcinogenic and genotoxic elements that cause lung cancer.[4] Overcoming nicotine dependence is often

extremely difficult. The Agency for Healthcare Research and Quality (formerly the Agency for Health Care Policy and Research [AHCPR]) developed a set of clinical smoking-cessation guidelines for helping nicotine-dependent patients and health care providers.[5] The six major elements of the guidelines include the following:

1. Clinicians must document the tobacco-use status of every patient.
2. Every patient using tobacco should be offered one or more of the effective smoking cessation treatments that are available.
3. Every patient using tobacco should be provided with at least one of the effective brief cessation interventions that are available.
4. More intense interventions are more effective than less intense interventions in producing long-term tobacco abstinence, reflecting the dose-response relationship between the intervention and its outcome.
5. One or more of the three treatment elements identified as being particularly effective should be included in smoking-cessation treatment:
 - a. Nicotine-replacement (e.g., nicotine patches and gum) or other evidence-based smoking cessation pharmacotherapy (e.g., varenicline or bupropion).
 - b. Social support from clinician in the form of encouragement and assistance.
 - c. Skills training and problem solving (cessation and abstinence techniques).
6. To be effective, health care systems must make institutional changes resulting in systematic identification of tobacco users and intervention with these patients at every visit.

Pharmacotherapy for smoking cessation

Many pharmacotherapies for smoking cessation, including nicotine replacement therapies (e.g., gum, patch, spray, lozenge, and inhaler) and other smoking cessation pharmacotherapies (e.g., varenicline and bupropion), result in statistically significant increases in smoking cessation rates compared with placebo. Based on a synthesis of the results of 110 randomized trials, nicotine replacement therapy treatments, alone or in combination, improve cessation rates over placebos after 6 months (relative risk, 1.58; 95% confidence interval, 1.50–1.66).[8] Since the AHCPR guidelines were published, additional evidence of the effectiveness of such pharmacotherapies for smoking cessation has been published.[9-11] The choice of therapy should be individualized based on a number of factors, including past experience, preference, and potential agent side effects. For more information on pharmacotherapy for smoking cessation, see [Cigarette Smoking: Health Risks and How to Quit](#).

Population-level interventions

In addition to individually focused cessation efforts, a number of tobacco control strategies at the community, state, and national level have been credited with reducing the prevalence of smoking. Strategies include the following:[12,13]

- Reducing minors' access to tobacco products.
- Disseminating effective school-based prevention curricula together with media strategies.
- Raising the cost of tobacco products by raising taxes.
- Using tobacco excise taxes to fund community-level interventions including mass media.
- Providing proven quitting strategies through health care organizations.
- Adopting smoke-free laws and policies.

Smoke-free workplace legislation

A review of more than 50 studies found that smoke-free workplace legislation was consistently associated with reduced secondhand smoke exposure, whether measured in reduced time of exposure (71%–100% reduction) or prevalence of persons exposed to secondhand smoke (22%–85% reduction), with particularly marked reductions among hospitality workers.[14] Smoke-free workplace legislation was associated with consistent and statistically significant reductions in levels of nicotine, dust, benzene, and particulate matter. Health indicators including respiratory systems, sensory symptoms, and hospital admissions were reported as outcomes in 25 studies. With respect to health outcomes, a consistent finding was reduced hospital admissions for cardiac events. Evidence suggested that smoke-free workplace legislation may also result in reduced prevalence of active cigarette smoking; for example, one study observed a 32% decreased smoking prevalence in a county that enacted smoke-free workplace legislation compared with a 2.8% decrease in nearby counties with no smoke-free workplace legislation.

Preventing Occupational Exposure to Lung Carcinogens

After cigarette smoking and exposure to secondhand smoke, occupational exposure to lung carcinogens, such as asbestos, arsenic, nickel, and chromium, is the most important contributor to the lung cancer burden. When occupational exposure to lung carcinogens are all considered together, 9% to 15% of all lung cancer deaths can be attributed to occupational exposure to lung carcinogens.[15] Reducing or eliminating workplace exposures to known lung carcinogens would be expected to result in a corresponding decrease in the risk of lung cancer. Consequently, the proportion of the lung cancer burden attributable to occupational exposures is declining over time in countries like the United States that have taken steps to protect the workforce from exposure to known lung carcinogens.

References

1. The Health Consequences of Smoking: A Report of the Surgeon General. U.S. Department of Health and Human Services, CDC, National Center for Chronic Disease Prevention and Health Promotion, Office on Smoking and Health, 2004. [Also available online](#). Last accessed April 9, 2025.
2. The Health Benefits of Smoking Cessation: a report of the Surgeon General. US Department of Health and Human Services, Public Health Service, Centers for Disease Control, Centers for Chronic Disease Prevention and Health Promotion, Office on Smoking and Health, DHHS Publ No (CDC) 90-8416, 1990.
3. Schottenfeld D, Fraumeni JF Jr, eds.: Cancer Epidemiology and Prevention. 2nd ed. Oxford University Press, 1996.
4. Cinciripini PM, Hecht SS, Henningfield JE, et al.: Tobacco addiction: implications for treatment and cancer prevention. J Natl Cancer Inst 89 (24): 1852-67, 1997. [\[PUBMED Abstract\]](#)
5. Fiore MC, Bailey WC, Cohen SJ, et al.: Smoking Cessation: Clinical Practice Guideline No 18. US Department of Health and Human Services, Public Health Service, Agency for Health Care Policy and Research, 1996. AHCPR Publ No 96-0692.
6. Tindle HA, Stevenson Duncan M, Greevy RA, et al.: Lifetime Smoking History and Risk of Lung Cancer: Results From the Framingham Heart Study. J Natl Cancer Inst 110 (11): 1201-1207, 2018. [\[PUBMED Abstract\]](#)
7. Greenlee RT, Murray T, Bolden S, et al.: Cancer statistics, 2000. CA Cancer J Clin 50 (1): 7-33, 2000 Jan-Feb. [\[PUBMED Abstract\]](#)
8. Silagy C, Lancaster T, Stead L, et al.: Nicotine replacement therapy for smoking cessation. Cochrane Database Syst Rev (3): CD000146, 2004. [\[PUBMED Abstract\]](#)
9. Hurt RD, Sachs DP, Glover ED, et al.: A comparison of sustained-release bupropion and placebo for smoking cessation. N Engl J Med 337 (17): 1195-202, 1997. [\[PUBMED Abstract\]](#)
10. Jorenby DE, Leischow SJ, Nides MA, et al.: A controlled trial of sustained-release bupropion, a nicotine patch, or both for smoking cessation. N Engl J Med 340 (9): 685-91, 1999. [\[PUBMED Abstract\]](#)
11. Hughes JR, Goldstein MG, Hurt RD, et al.: Recent advances in the pharmacotherapy of smoking. JAMA 281 (1): 72-6, 1999. [\[PUBMED Abstract\]](#)
12. Wingo PA, Ries LA, Giovino GA, et al.: Annual report to the nation on the status of cancer, 1973-1996, with a special section on lung cancer and tobacco smoking. J Natl Cancer Inst 91 (8): 675-90, 1999. [\[PUBMED Abstract\]](#)
13. Koh HK: The end of the "tobacco and cancer" century. J Natl Cancer Inst 91 (8): 660-1, 1999. [\[PUBMED Abstract\]](#)

14. Callinan JE, Clarke A, Doherty K, et al.: Legislative smoking bans for reducing secondhand smoke exposure, smoking prevalence and tobacco consumption. *Cochrane Database Syst Rev* (4): CD005992, 2010. [\[PUBMED Abstract\]](#)
15. Alberg AJ, Ford JG, Samet JM, et al.: Epidemiology of lung cancer: ACCP evidence-based clinical practice guidelines (2nd edition). *Chest* 132 (3 Suppl): 29S-55S, 2007. [\[PUBMED Abstract\]](#)

Interventions Associated With Increased Risk of Lung Cancer

Beta-Carotene Supplementation in Smokers

Results of the National Cancer Institute (NCI) Alpha-Tocopherol, Beta-Carotene Cancer Prevention (ATBC) trial were first published in 1994.[\[1\]](#) This trial included 29,133 Finnish male chronic smokers aged 50 to 69 years in a 2 × 2 factorial design of alpha-tocopherol (50 mg/day) and beta-carotene (20 mg/day). Participants were randomly assigned to one of the following four groups for 5 to 8 years: beta-carotene alone, alpha-tocopherol alone, beta-carotene plus alpha-tocopherol, or placebo. Participants who received beta-carotene (alone or with alpha-tocopherol) had a higher incidence of lung cancer (relative risk [RR], 1.18; 95% confidence interval [CI], 1.03–1.36) and higher total mortality (RR, 1.08; 95% CI, 1.01–1.16). This effect appeared to be associated with heavier smoking (one or more packs/day) and alcohol intake (at least one drink/day).[\[2\]](#) Supplementation with alpha-tocopherol produced no overall effect on lung cancer (RR, 0.99; 95% CI, 0.87–1.13).

In 1996, the results of the U.S. Beta-Carotene and Retinol Efficacy Trial (CARET) were published.[\[3\]](#) This multicenter trial involved 18,314 smokers, former smokers, and asbestos-exposed workers who were randomly assigned to beta-carotene (at a higher dose than the ATBC trial, 30 mg/day) plus retinyl palmitate (25,000 IU/day) or placebo. The primary endpoint was lung cancer incidence. The trial was terminated early by the Data Monitoring Committee and NCI because its results confirmed the ATBC finding of a harmful effect of beta-carotene over that of placebo, which increased lung cancer incidence (RR, 1.28; 95% CI, 1.04–1.57) and total mortality (RR, 1.17; 95% CI, 1.03–1.33). In a follow-up study of CARET participants after the intervention discontinued, these effects attenuated for a period of time. After 6 years of postintervention follow-up, the postintervention RR for lung cancer incidence was 1.12 (95% CI, 0.97–1.31) and for total mortality was 1.08 (95% CI, 0.99–1.71). During the postintervention phase a larger RR among women, rather than men, emerged for both outcomes in subgroup analyses; the reason for this observation, if reliable, is not known.[\[4\]](#)

The overall findings from the ATBC [\[1,2\]](#) and CARET [\[3,5\]](#) studies, which include over 47,000 participants, demonstrated that pharmacological doses of beta-carotene increase lung cancer

risk in relatively high-intensity smokers. The mechanism of this adverse effect is not known. Lung cancer risks were not increased in subsets of moderate-intensity smokers (less than a pack per day) in the ATBC study, or in former smokers in the CARET study. Evidence from other studies, such as the Physicians' Health Study,[6] does not indicate that beta-carotene supplementation increases lung cancer risk in nonsmokers. Subsequent analyses of participants in these trials and cohorts suggest that the beneficial outcomes associated with high beta-carotene plasma levels may be due to increased dietary intake of fruits and vegetables. These findings show the importance of randomized controlled trials to confirm epidemiological studies.

References

1. The effect of vitamin E and beta carotene on the incidence of lung cancer and other cancers in male smokers. The Alpha-Tocopherol, Beta Carotene Cancer Prevention Study Group. *N Engl J Med* 330 (15): 1029-35, 1994. [[PUBMED Abstract](#)]
2. Albanes D, Heinonen OP, Taylor PR, et al.: Alpha-Tocopherol and beta-carotene supplements and lung cancer incidence in the alpha-tocopherol, beta-carotene cancer prevention study: effects of base-line characteristics and study compliance. *J Natl Cancer Inst* 88 (21): 1560-70, 1996. [[PUBMED Abstract](#)]
3. Omenn GS, Goodman GE, Thornquist MD, et al.: Effects of a combination of beta carotene and vitamin A on lung cancer and cardiovascular disease. *N Engl J Med* 334 (18): 1150-5, 1996. [[PUBMED Abstract](#)]
4. Goodman GE, Thornquist MD, Balmes J, et al.: The Beta-Carotene and Retinol Efficacy Trial: incidence of lung cancer and cardiovascular disease mortality during 6-year follow-up after stopping beta-carotene and retinol supplements. *J Natl Cancer Inst* 96 (23): 1743-50, 2004. [[PUBMED Abstract](#)]
5. Omenn GS, Goodman GE, Thornquist MD, et al.: Risk factors for lung cancer and for intervention effects in CARET, the Beta-Carotene and Retinol Efficacy Trial. *J Natl Cancer Inst* 88 (21): 1550-9, 1996. [[PUBMED Abstract](#)]
6. Hennekens CH, Buring JE, Manson JE, et al.: Lack of effect of long-term supplementation with beta carotene on the incidence of malignant neoplasms and cardiovascular disease. *N Engl J Med* 334 (18): 1145-9, 1996. [[PUBMED Abstract](#)]

Interventions With Adequate Evidence That They Do Not Reduce Risk

Chemoprevention

Studies have examined whether it is possible to prevent cancer development in the lung using chemopreventive agents. Chemoprevention is defined as the use of specific natural or synthetic chemical agents to reverse, suppress, or prevent carcinogenesis before the development of invasive malignancy. So far, agents tested for efficacy in lung cancer chemoprevention have been micronutrients, such as beta-carotene and vitamin E.

Beta-carotene supplementation in nonsmokers

Two other randomized controlled trials of beta-carotene were carried out in populations that were not at excess risk of lung cancer. The Physicians' Health Study was designed to study the effects of beta-carotene and aspirin in cancer and cardiovascular disease. The study was a randomized, double-blind, placebo-controlled trial begun in 1982 involving 22,071 male physicians aged 40 to 84 years. After 12 years of follow-up, beta-carotene was not associated with overall risk of cancer (relative risk [RR], 0.98) or lung cancer among current (11% of study population) or former (39% of study population) smokers.[1]

In the Women's Health Study (WHS) approximately 40,000 female health professionals were randomly assigned to 50 mg beta carotene on alternate days or placebo. After a median of 2.1 years of beta-carotene treatment and 2 additional years of follow-up, there was no evidence that beta-carotene protected against lung cancer, as there were more lung cancer cases observed in the beta-carotene (n = 30) than placebo (n = 21) group.[2] The strong evidence from rigorous randomized, placebo-controlled trials clearly indicated that beta-carotene supplementation does not lower the risk of lung cancer in populations that are not high-risk for lung cancer.

Vitamin E supplementation

The Heart Outcomes Prevention Evaluation (HOPE) trial began in 1993 and continued follow-up as the HOPE-The Ongoing Outcomes (HOPE-TOO) through 2003. In this randomized, placebo-controlled trial, patients aged 55 years or older with vascular disease or diabetes were assigned to 400 IU vitamin E or placebo. With a median follow-up of 7 years, the group randomly assigned to vitamin E had a significantly lower lung cancer incidence rate (1.4%) than the placebo group (2.0%) (RR, 0.72; 95% confidence interval [CI], 0.53–0.98).[3] However, the protective association between vitamin E supplements and lung cancer in the HOPE-TOO study needs to be interpreted in the context of evidence from other randomized trials. In the National Cancer Institute Alpha-Tocopherol, Beta-Carotene Cancer Prevention (ATBC) study, supplementation with alpha-tocopherol produced no overall effect on lung cancer (RR, 0.99; 95% CI, 0.87–1.13). In the WHS of 40,000 female health professionals, using 600 IU of vitamin E every other day showed no evidence of protection against lung cancer in women (RR, 1.09; 95% CI, 0.83–1.44).[4] The Medical Research Council/British Heart Foundation Heart Protection Study (HPS) is a randomized, placebo-controlled trial to test antioxidant vitamin supplementation with vitamin E, vitamin C, and beta-carotene among 20,536 United Kingdom

adults with coronary disease, other occlusive arterial disease, or diabetes. The trial began recruitment in 1994, and as of the 2001 follow-up the results showed a slightly higher rate of lung cancer in the vitamin group compared with the placebo group (1.6% vs. 1.4%, respectively).[5]

Looking at the vitamin E results for the ATBC, HPS, and HOPE-TOO studies combined, the summary odds ratio was 0.97 (95% CI, 0.87–1.08),[3] and adding the results from the WHS to this would bring the association even closer to the null. The combined evidence for vitamin E supplementation thus continues to be consistent with no effect on lung cancer risk.

References

1. Hennekens CH, Buring JE, Manson JE, et al.: Lack of effect of long-term supplementation with beta carotene on the incidence of malignant neoplasms and cardiovascular disease. *N Engl J Med* 334 (18): 1145-9, 1996. [\[PUBMED Abstract\]](#)
2. Lee IM, Cook NR, Manson JE, et al.: Beta-carotene supplementation and incidence of cancer and cardiovascular disease: the Women's Health Study. *J Natl Cancer Inst* 91 (24): 2102-6, 1999. [\[PUBMED Abstract\]](#)
3. Lonn E, Bosch J, Yusuf S, et al.: Effects of long-term vitamin E supplementation on cardiovascular events and cancer: a randomized controlled trial. *JAMA* 293 (11): 1338-47, 2005. [\[PUBMED Abstract\]](#)
4. Lee IM, Cook NR, Gaziano JM, et al.: Vitamin E in the primary prevention of cardiovascular disease and cancer: the Women's Health Study: a randomized controlled trial. *JAMA* 294 (1): 56-65, 2005. [\[PUBMED Abstract\]](#)
5. Heart Protection Study Collaborative Group: MRC/BHF Heart Protection Study of antioxidant vitamin supplementation in 20,536 high-risk individuals: a randomised placebo-controlled trial. *Lancet* 360 (9326): 23-33, 2002. [\[PUBMED Abstract\]](#)

Latest Updates to This Summary (04/17/2025)

The PDQ cancer information summaries are reviewed regularly and updated as new information becomes available. This section describes the latest changes made to this summary as of the date above.

Incidence and Mortality

Updated [statistics](#) with estimated new cases and deaths for 2025 (cited American Cancer Society as reference 1).

This summary is written and maintained by the [PDQ Screening and Prevention Editorial Board](#), which is editorially independent of NCI. The summary reflects an independent review of the literature and does not represent a policy statement of NCI or NIH. More information about summary policies and the role of the PDQ Editorial Boards in maintaining the PDQ summaries can be found on the [About This PDQ Summary](#) and [PDQ® Cancer Information for Health Professionals](#) pages.

About This PDQ Summary

Purpose of This Summary

This PDQ cancer information summary for health professionals provides comprehensive, peer-reviewed, evidence-based information about lung cancer prevention. It is intended as a resource to inform and assist clinicians in the care of their patients. It does not provide formal guidelines or recommendations for making health care decisions.

Reviewers and Updates

This summary is reviewed regularly and updated as necessary by the [PDQ Screening and Prevention Editorial Board](#), which is editorially independent of the National Cancer Institute (NCI). The summary reflects an independent review of the literature and does not represent a policy statement of NCI or the National Institutes of Health (NIH).

Board members review recently published articles each month to determine whether an article should:

- be discussed at a meeting,
- be cited with text, or
- replace or update an existing article that is already cited.

Changes to the summaries are made through a consensus process in which Board members evaluate the strength of the evidence in the published articles and determine how the article should be included in the summary.

Any comments or questions about the summary content should be submitted to Cancer.gov through the NCI website's [Email Us](#). Do not contact the individual Board Members with questions or comments about the summaries. Board members will not respond to individual inquiries.

Levels of Evidence

Some of the reference citations in this summary are accompanied by a level-of-evidence designation. These designations are intended to help readers assess the strength of the evidence supporting the use of specific interventions or approaches. The PDQ Screening and Prevention Editorial Board uses a [formal evidence ranking system](#) in developing its level-of-evidence designations.

Permission to Use This Summary

PDQ is a registered trademark. Although the content of PDQ documents can be used freely as text, it cannot be identified as an NCI PDQ cancer information summary unless it is presented in its entirety and is regularly updated. However, an author would be permitted to write a sentence such as “NCI’s PDQ cancer information summary about breast cancer prevention states the risks succinctly: [include excerpt from the summary].”

The preferred citation for this PDQ summary is:

PDQ® Screening and Prevention Editorial Board. PDQ Lung Cancer Prevention. Bethesda, MD: National Cancer Institute. Updated <MM/DD/YYYY>. Available at: <https://www.cancer.gov/types/lung/hp/lung-prevention-pdq>. Accessed <MM/DD/YYYY>. [PMID: 26389452]

Images in this summary are used with permission of the author(s), artist, and/or publisher for use within the PDQ summaries only. Permission to use images outside the context of PDQ information must be obtained from the owner(s) and cannot be granted by the National Cancer Institute. Information about using the illustrations in this summary, along with many other cancer-related images, is available in [Visuals Online](#), a collection of over 2,000 scientific images.

Disclaimer

The information in these summaries should not be used as a basis for insurance reimbursement determinations. More information on insurance coverage is available on Cancer.gov on the [Managing Cancer Care](#) page.

Contact Us

More information about contacting us or receiving help with the Cancer.gov website can be found on our [Contact Us for Help](#) page. Questions can also be submitted to Cancer.gov through the website’s [Email Us](#).

Updated: April 17, 2025

If you would like to reproduce some or all of this content, see [Reuse of NCI Information](#) for guidance about copyright and permissions. In the case of permitted digital reproduction, please credit the National Cancer Institute as the source and link to the original NCI product using the original product's title; e.g., "Lung Cancer Prevention (PDQ®)–Health Professional Version was originally published by the National Cancer Institute."



Want to use this content on your website or other digital platform? Our [syndication services page](#) shows you how.